

Probability Mapping Based Artifact Detection and Wavelet Denoising based Artifact Removal from Scalp EEG for BCI Applications

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Abstract—In EEG-based Brain-Computer Interface (BCI) applications, the EEG recording is often contaminated by different types of artifacts that can misinterpret the BCI output. Automatic detection and removal of such offending artifacts from EEG for online processing pose a great challenge. In this paper, we present a novel method that can map the artifact probability of an EEG epoch based on four statistical measures: *entropy*, *kurtosis*, *skewness* and *Periodic Waveform Index (PWI)*. Then a removal method is adopted based on stationary wavelet transform that can be applied to the epochs by setting a particular probability threshold from the user. This epoch by epoch preprocessing would allow the user to tune the threshold parameters after some initial training with the same EEG recordings and eventually can be applied to both offline and online processing. Experimental results with both simulated and real EEG data prove the efficacy of the method that it can reliably trace the artifactual epoch with reasonable accuracy and eventually reduces the artifacts from EEG with very little distortion to the signal of interest. Further testing with EEG datasets for BCI experiments also shows that artifact removal can significantly enhance the BCI performance in both motor-imagery (MI) and event related potential (ERP) based BCI applications.

Keywords- EEG; Artifact Detection; Artifact Removal; Stationary Wavelet Transform (SWT); Brain-Computer Interface (BCI).

I. INTRODUCTION

Brain-Computer Interface (BCI) is a promising technique for establishing a direct link between human brain and an external computerized device bypassing the normal pathway when it is not functional due to any brain/spinal cord related injury or diseases [1]. BCI allows severely disabled persons to communicate with outside world by controlling certain computerized device such as computer, wheelchair, neural prosthetics, etc. In addition to that it is also used as a rehabilitation tool for stroke patients and people with similar needs. Although, there are different brain recording techniques (starting from non-invasive to invasive ones) to measure the electrical activity of brain to process and use in BCI applications, scalp EEG is the most popular among them in BCI research due to its mainly non-invasive nature with other attractive features such as fine temporal resolution, simple to use, portability and lower cost [2]. However, scalp EEG is very much prone to undesirable artifacts that come from non-cerebral origin. The artifacts often severely

contaminate the EEG recordings and modify the shape of a particular neurological event that drives the BCI system affecting the accuracy of the BCI performance. For example, artifacts can mistakenly cause an unintentional control of the BCI device [3].

Therefore, handling of such offending artifacts is critical in BCI research for satisfactory performance and consequently many techniques have been developed to get rid of artifacts that create misinterpretation in the signal analysis. Recent advances in signal processing techniques have now allowed the researchers to design automatic artifact identification and removal algorithms to be able to implement online for real world BCI applications.

There have been several methods proposed in the literature to detect and remove artifacts particularly for BCI applications. One of the most common is the use of Blind Source Separation based algorithm (e.g. Independent Component Analysis or ICA [4], [5], [6]) which separates multi-channel EEG recordings into independent or maximally uncorrelated sources and then rejecting the artifact source before reconstructing the signals. However, ICA is an offline technique and often requires manual intervention to identify the artifactual source (i.e. Independent Components or ICs). Moreover, it cannot operate on few or single-channel datasets thus not suitable for portable ambulatory BCI applications. Another popular method to remove physiological artifacts is linear regression (e.g. using adaptive filter [5], [7]) with the availability of a dedicated reference artifact channel such as EOG or ECG. In many BCI applications, such reference channel may not be available thus not possible to use regression technique. In recent times, two methods namely wavelet transform [8], [9], [10] and empirical mode decomposition (EMD) [11], [10], that are suitable for nonlinear and non-stationary biomedical signal processing (e.g. EEG), have been introduced to detect and remove artifacts from EEG. Among these two, wavelet based techniques combining with another technique such as ICA/CCA [8], [9] or neural network [5], [7] have been very popular and stated to have decent performance in artifact removal. On the other hand, EMD is more computationally expensive and still an empirical method that doesn't have complete theoretical background.

In this paper, we propose a unique artifact detection method based on artifact probability mapping which quantifies an epoch by a relative probability of being artifactual. This is achieved by considering the typical

artifact characteristics in contrast to the background EEG rhythms with the help of four statistical measures namely entropy, kurtosis, skewness and periodicity as described in detail in the later section of this article. This unique probability mapping will allow the user to decide whether to correct the epoch or to remain as it is by choosing an appropriate probability threshold. Subsequently a removal method is also proposed which is based on stationary wavelet transform based denoising and relies on the desired spectral band of EEG rhythms in contrast to spectral bands of different artifacts particularly for brain computer interface (BCI) applications. The proposed method is demonstrated with both real and synthesized database and extensive quantitative measures show the efficacy of this method with obtained satisfactory results. The effect of artifact removal on real BCI database has also been demonstrated to show that it can substantially improve the classifier accuracy in BCI experiments.

II. PROPOSED METHOD

The proposed system architecture is given in figure 1.

A. Artifact Detection: Probability Mapping

The proposed artifact probability mapping is similar to the idea of spike probability mapping described in [12] where the authors proposed a probability mapping for separation/identification of neural action potential or spike from background noise for in-vivo neural recordings. Here we attempt to separate artifacts from background EEG rhythms by representing each epoch of 1-sec duration by the relative probability of artifact contamination. This probability of artifact contamination depends on four measures as:

- Entropy means randomness of samples in an epoch
- Kurtosis representing peakedness of samples in an epoch
- Skewness representing symmetry of distribution of the samples in an epoch
- Periodic waveform index (PWI) that represents the periodicity in an epoch

1) *Entropy, H*: Entropy, a measure of uncertainty of information content, of a discrete random variable x with possible values $x_1, \dots, x_n$, can be found as below:

$$H(x) = E[-\ln(P(x))]. \quad (1)$$

where E and $P(x)$ are the expected value operator and probability mass function of x respectively.

2) *Kurtosis, Kr*: It is a measure of "peakedness" of the probability distribution function and can be defined for a real-valued random variable x as:

$$Kr[x] = \frac{\mu^4}{\sigma^4}. \quad (2)$$

where μ and σ are the mean and standard deviation of random variable x respectively.

3) *Skewness, γ_1* : How much asymmetry presence about its mean for the probability distribution of a random variable, x , is measured through skewness and calculated as:

$$\gamma_1 = \frac{\mu^3}{\sigma^3}. \quad (3)$$

4) *Periodic Waveform Index, PWI*: Periodic Waveform Index is the measure of the rhythmicity or periodicity pattern in a signal x proposed by [13]. Suppose the total harmonic energy at time point n is $E_\tau[n]$ and the signal energy corresponding to a cycle duration τ is $N_\tau[n]$ then PWI is defined as

$$PWI[n] \triangleq \frac{E_\tau[n]}{N_\tau[n]}. \quad (4)$$

$$N_\tau[n] \triangleq \frac{1}{\tau} \sum_{n'=-\infty}^{\infty} (x[n']\psi[n'-n])^2 \quad (5)$$

$$E_\tau[n] \triangleq \sum_{k>0} \left| \frac{1}{\tau} \sum_{n'=-\infty}^{\infty} (x[n']\psi[n'-n]) \exp(-2i\pi \frac{kn'}{\tau}) \right|^2 \quad (6)$$

Here ψ is a window of bounded energy centered around 0 of length $a\tau$ and $\hat{\tau} = \arg. \max_{\tau} E_\tau$.

5) *Probability of Artifact, Pr*: Probability of an epoch being artifact-contaminated depends on four above mentioned measures. By studying the artifact characteristics [14], [15], we found that artifact tends to be more random (i.e. lower entropy values) with higher amplitude (i.e. higher kurtosis) than background EEG, more likely to be transient (i.e. lower PWI) than rhythmic (i.e. unlike EEG rhythms) and artifact makes the data distribution to be asymmetrical e.g. longer tail on either side of data histogram (i.e. higher skewness). Therefore, keeping in mind of these facts, we propose an equation as follows to measure the probability of being artifactual for an epoch:

$$Pr = \frac{[k_H(1 - \hat{H})] + [k_{Kr}\hat{Kr}] + [k_{\gamma_1}\hat{\gamma}_1] + [k_{PWI}(1 - \hat{PWI})]}{4} \quad (7)$$

Where $k_H, k_{Kr}, k_{\gamma_1}$, and k_{PWI} are the weights of the measures $\hat{H}, \hat{Kr}, \hat{\gamma}_1$, and $\hat{PWI}$ respectively that are normalized by their corresponding maximum values.

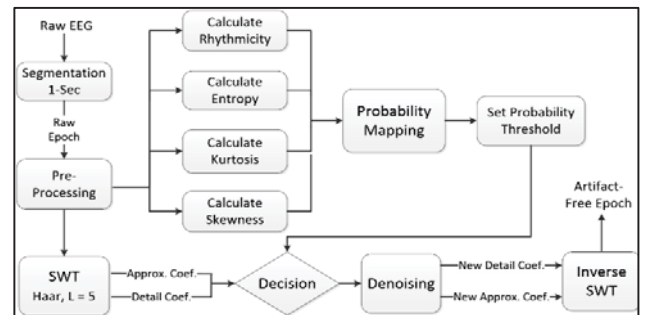


Figure 1. Proposed structure to create probability mapping and subsequently to remove artifacts from EEG

B. SWT-Based Artifact Removal

Once the artifacts are detected, to remove them we have followed similar process of Stationary Wavelet Transform based denoising as described in [15] in order to get artifact free EEG data. We briefly describe the removal process in this sub-section which has four major steps as follows:

1) *Preprocessing*: Assuming $x_{raw}(n)$ as the raw EEG signal which is sampled at F_s Hz where $n = \{0,1,2,\dots\}$ being the discrete-time index. Considering that the power-line interference (50/60 Hz and its harmonics) as well as the DC offset have already been removed prior to this stage. In the preprocessing, the incoming raw EEG sequence is first of all divided into non-overlapping epochs with window size of N .

2) *Wavelet Decomposition*: The purpose of this stage is to decompose and analyze the EEG epochs in wavelet domain for possible identification of the presence of artifactual components in the later stage. To this end, stationary wavelet transform (SWT) is applied on the epochs $\{x_j\}_{j \geq 1}$ with level-5 decomposition by Haar as basis wavelet which results in final approximate $a_{j,5}$ and detail coefficients $d_{j,1}, d_{j,2}, \dots, d_{j,5}$. The choice for level of decomposition is primarily motivated from the bandwidth of EEG signal (i.e. 0.05 - 128 Hz) and the frequency bands of EEG rhythms relevant to BCI studies such as mu (7 - 13Hz) and beta (14 - 30Hz) for MI-based BCI [16]; delta (< 4Hz) for ERP-based BCI; e.g. P300 [17]; and SSVEP (12-18 Hz) [18] in order to have enough no. of frequency sub-bands to make decisions on where to denoise carefully and where not. Table I shows the frequency sub-bands of the decomposed wavelet coefficients after 5-level SWT is performed on the raw EEG data. It is clear that D_3 , D_4 and A_5 correspond EEG rhythms beta, mu and delta respectively and hence during denoising process, very careful approach is required to handle these coefficients. The other three coefficients, i.e. D_1 , D_2 and D_5 can be denoised directly and fully without any further consideration as most likely they contain artifacts and non-relevant EEG rhythms in such BCI applications.

TABLE I. ILLUSTRATION OF SWT COEFFICIENTS IN RELATION TO EEG RHYTHMS, ARTIFACTS AND RELEVANT BCI STUDIES IN THEIR CORRESPONDING FREQUENCY BANDS.

SWT Coeff.	F_s (Hz)	D_1	D_2	D_3	D_4	D_5	A_5
Freq. Band (Hz)	256	64 - 128	32 - 64	16 - 32	8 - 16	4 - 8	0 - 4
EEG Rhythms			Gamma	Beta	Mu Alpha	Theta	Delta
Relevant BCI Studies				Motor Imagery BCI SSVEP		ERP (P300)	
Artifact Type			EMG (Muscle Artifacts) Electrode Pop			Cardiac EOG Blink Movement	

3) *Denoising*: Non-negative garrote shrinkage function is used during denoising as it's a good trade-off between soft and hard threshold functions in terms of removal performance and signal distortion [19]. In order to denoise the critical coefficients $d_{j,3} - d_{j,4}$ for MI-BCI and $a_{j,5}$ for P300-based BCI, and $d_{j,4}$ for SSVEP-based BCI, we have used modified universal threshold reported by [14]:

$$t'_{i,\ell} = K\alpha_{i,\ell}\sqrt{2\ln N}, \quad (8)$$

where in (8) N is the length of epoch and $\alpha_{i,\ell}$ is the estimated noise variance for $w_{i,\ell}$.

4) *Decision*: Depending upon this stage, the decision of whether an epoch is to be denoised or not is made from the probability threshold set by user. In addition, once the decision of denoising is made, then how carefully that particular epoch is to be denoised to remove artifacts without affecting the signal of interest, is also decided in this stage. The first step of this stage is to find whether the epoch is truly artifactual based on response from both probability mapping and from measuring the modified universal threshold for each level of decomposed wavelet coefficients: if both of them are affirmative, then the epoch is decided to denoise fully. Otherwise we let the epoch as it is. The parameters are chosen as we discuss in the previous part of Wavelet Decomposition. As we denoise primarily based on the user's set probability threshold, some epochs still being found artifactual from both probability mapping and wavelet coefficients, they remain untouched and therefore we pay the penalty of less amount of artifact removal. The pseudo code for decision stage is provided in Table II.

5) *Reconstruction*: During the final stage of signal reconstruction, taking the decision from the *Decision* stage, either full thresholding or partial (or not at all) thresholding is applied to obtain a new set of coefficients ($D'_1 - D'_5$ and A'_5). Finally, inverse SWT is applied on the newly formed coefficients to reconstruct the EEG epochs. Therefore, a new sequence of EEG known as *Reconstructed EEG* is obtained.

III. METHODS/DATA COLLECTION

A. Data Collection

Real EEG recordings are downloaded from BCI competition-IV Scalp EEG Database: dataset-1 from [2], dataset-2a and dataset-2b from [1], [20], and [21]. The dataset-1 has 64 EEG channels (0.05-200 Hz) and sampled at 1000 Hz recorded from 7 subjects for MI-BCI study. Dataset-2a has 22 EEG channels (0.5-100 Hz) with 3 EOG channels recorded from 9 subjects having 4 classes whereas dataset-2b has 3 bipolar EEG channels + 3 EOG channels collected from 9 subjects having 2 classes for MI-based BCI experiment. Both of them were sampled at 250 Hz.

B. Data Synthesis

Two signal components have been simulated: artifacts and EEG background activities (aka EEG rhythm) and the linear combination of them makes a new dataset termed as 'Artifactual EEG'. The individual artifact templates during chewing, swallowing, eye blinks, subject movements, talking, etc. are also simulated and added randomly with the background EEG. The simulation of background EEG has been performed according to the classical Event Related Potentials (ERP) theory as per given in [22]. The MATLAB source code used to simulate such background EEG is available online for free download [23].

TABLE II. PSEUDO-CODE FOR THE DECISION STAGE DURING DENOISING SWT COEFFICIENTS

Decision Making	Remarks/Comment
<i>If (P300 ERP Study)</i> $k_{A_5} = 1,$ $k_{D_5} = 1$ $k_{D_i} = 0.75, i = 2,3,4$ $k_{D_1} = 0.5$ <i>End</i>	Denoise D_5, A_5 carefully
<i>If (MI Study)</i> $k_{A_5} = 0.5, k_{D_5} = 0.75$ $k_{D_i} = 1, i = 3,4$ $k_{D_2} = 0.75, k_{D_1} = 0.5$ <i>End</i>	Denoise D_2-D_4 carefully
<i>If (SSVEP Study)</i> $k_{A_5} = 0.5, k_{D_5} = 0.75$ $k_{D_4} = 1$ $k_{D_3} = 0.75, k_{D_2} = 0.75, k_{D_1} = 0.5$ <i>End</i>	Denoise D_4 carefully

IV. RESULTS AND DISCUSSION

The performance of the proposed algorithm on the artifact removal is assessed both in terms of the amount of artifact reduction and the amount of signal using the efficiency metrics proposed and described in detailed in [15], [24]. In order to have a fair evaluation, the amount and duration of artifacts present in the signals are also considered. In order to evaluate the BCI performance, ROC curve parameters are calculated.

A. Qualitative Evaluation

1) *Real Data*: The proposed removal algorithm is tested on real EEG recordings from a healthy subject with labeled artifacts during chewing and eye-blink. The artifact removal result in time-domain is shown in Fig. 2(a) for qualitative evaluation which shows that the removal of both types of artifacts without distorting the background EEG rhythms in the non-artifactual region is quite satisfactory.

2) *Simulated*: The artifact removal performance for a fully simulated EEG signal is illustrated in figure 3 where both signal components (artifacts and background EEG rhythms) are simulated. As it's seen that the synthesized artifactual EEG is heavily contaminated with different types of artifacts and its corresponding probability mapping is also shown. Once the artifact removal is performed, the probability of being artifactual for previously contaminated epochs have been reduced significantly as it is obvious from the lower part of the plot. Please note that it's not possible to remove all the artifacts completely (i.e. 100% λ), but the purpose is to reduce artifact contamination as much as possible without distorting the signal of interest.

B. Quantitative Evaluation

1) *Artifact Removal*: Table III presents the quantitative metrics of artifact removal with respect to the strength of artifacts, i.e. different artifact SNR SNR_{Art} and different artifact duration ΔT_{Art} . The upper part (i.e. row) of the table represents the quantitative values without considering the set probability threshold while the lower row represents after considering the probability mapping and its probability threshold.

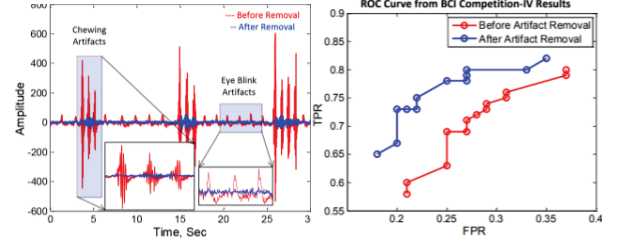


Figure 2. (a) The removal result after the proposed algorithm is applied on our recorded EEG. (b) ROC curve plotted for signals before and after artifact removal from the BCI output using LDA Classifier and Windowed Means as features from BCILAB tool.

2) *BCI Performance*: We have used BCILAB [25] which is an open source toolbox compatible with EEGLAB [26] for BCI performance evaluation. The BCI-IV competition dataset (motor-imagery experimental BCI) was used to observe the classification results for both before and after artifact removal signals. Windowed means features with LDA classifier is used for such evaluation. The test results is shown in terms of ROC curve in figure 2(b) suggests that BCI classifier outputs get better after artifact removal is applied and hence it can be used to improve the BCI experience for the users.

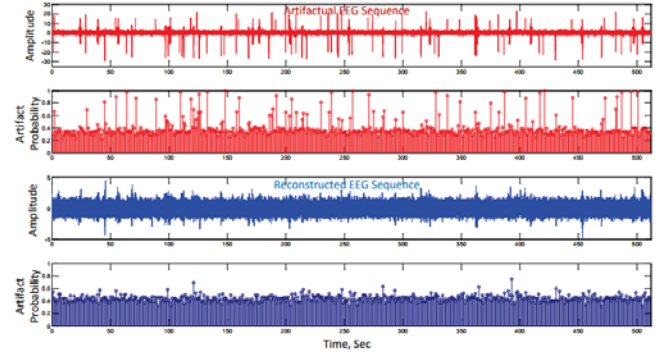


Figure 3. Fully simulated EEG data sequence with different types of artifacts before and after artifact removal (upper plot) and their corresponding probability map (lower plot). The time window duration in this case is chosen as 1 sec

V. CONCLUSION

The presented probability mapping of artifact for each EEG epoch will allow the user to choose whether to denoise any epoch or not and hence providing the flexibility in artifact handling for different BCI applications. Also the use of wavelet transform in artifact removal provides the use of single-channel BCI applications that has recently been popular [27] and also eliminate the requirement of a reference artifact channel. The scope of employing the proposed artifact removal for both ERP-based and MI-based BCI application is another advantage of this presented work. Although the current method has been tested for offline analysis, but since it is consistent with the typical epoch-by-epoch handling of BCI processing, we expect this method to be able to implement online after necessary modification.

TABLE III. QUANTITATIVE METRICS OF REMOVAL RESULTS FOR CORRESPONDING ARTIFACT SNR (SNR_{Art}) AND ARTIFACT DURATION RATIO

SNR_{Art} (dB)	$T_{Art}(\%)$	Λ (%)	ΔSNR (dB)	$\Delta RMSE$ (%)	ΔPSD_{dis} (%)	$\Delta Corr$ (%)	ΔCoh (%)	$\Delta SNDR$ (dB)
11.39 (1.95)	20.15 (3.66)	43.37 (8.09)	12.85 (1.53)	76.87 (4.21)	96.64 (0.05)	126.88 (39.89)	51.45 (20.51)	11.26 (1.84)
SNR_{Art} (dB)	$T_{Art}(\%)$	Λ (%)	ΔSNR (dB)	$\Delta RMSE$ (%)	ΔPSD_{dis} (%)	$\Delta Corr$ (%)	ΔCoh (%)	$\Delta SNDR$ (dB)
15.93 (2.17)	14.06 (2.39)	48.45 (6.88)	14.84 (1.67)	81.77 (3.57)	99.85 (0.14)	109.73 (34.82)	8.22 (4.79)	16.54 (1.58)

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